Notes

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1 Categories

Definition 1.1. A **category** $\mathcal{C}$ consists of

1. a collection of objects $\mathrm{ob}(\mathcal{C})$
2. for each $A, B \in \mathrm{ob}(\mathcal{C})$ a collection of morphisms $\mathrm{Hom}_{\mathcal{C}}(A, B)$ called the hom-set of A and B
   We will write $f : A \to B$ or $A \xrightarrow{f} B$ for $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$
3. for each $A, B, C \in \mathrm{ob}(\mathcal{C})$ an assignment $\circ : \mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C)$ such that for morphisms $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$ we have $h \circ (g \circ f) = (h \circ g) \circ f$
4. for each object $A \in \mathrm{ob}(\mathcal{C})$ we have a morphism $1_A : A \to A$ satisfying for all morphisms $X \xrightarrow{f} A \xrightarrow{g} Y$ the identity
   $$1_A \circ f = f \qquad g \circ 1_A = g \tag{1}$$
   which we will call the identity morphism.

Remark. From (1) it follows that the identity morphism 1_A is unique, since for $1_A, \hat{1}_A$ we have

$$1_A = 1_A \circ \hat{1}_A = \hat{1}_A$$

Definition 1.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We will say that $\mathcal{C}$ is a **subcategory** of $\mathcal{D}$, if $\mathrm{ob}(\mathcal{C})$ is a subcollection of $\mathrm{ob}(\mathcal{D})$, for each $A, B \in \mathrm{ob}(\mathcal{C})$ the hom-set $\mathrm{Hom}_{\mathcal{C}}(A, B)$ is a subcollection of $\mathrm{Hom}_{\mathcal{D}}(A, B)$, the identity morphisms in $\mathcal{C}$ are the identity morphisms in $\mathcal{D}$ and the composition $\circ$ in $\mathcal{C}$ is the restriction of composition in $\mathcal{D}$. If $\forall A, B \in \mathrm{ob}(\mathcal{C}) : \mathrm{Hom}_{\mathcal{C}}(A, B) = \mathrm{Hom}_{\mathcal{D}}(A, B)$ then we call $\mathcal{C}$ a **full subcategory** of $\mathcal{D}$.

Definition 1.3. A category $\mathcal{C}$ is called **locally small** if all of the hom-sets in $\mathcal{C}$ are sets. A category $\mathcal{C}$ is **small** if it is locally small and $\mathrm{ob}(\mathcal{C})$ is a set.

Definition 1.4. For a category $\mathcal{C}$ we define the **opposite category** $\mathcal{C}^{\mathrm{op}}$ by

$$\mathrm{ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{ob}(\mathcal{C})$$

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A)$$

For $A, B, C \in \mathrm{ob}(\mathcal{C}^{\mathrm{op}})$ and morphisms $A \xrightarrow{f} B \xrightarrow{g} C$ we'll let $g \circ_{\mathrm{op}} f = f \circ g$ and define the identity morphisms of $\mathcal{C}^{\mathrm{op}}$ as the identity morphisms of $\mathcal{C}$.

The opposite category is a category since

$$h \circ_{\mathrm{op}} (g \circ_{\mathrm{op}} f) = (g \circ_{\mathrm{op}} f) \circ h = (f \circ g) \circ h = f \circ (g \circ h) = (g \circ h) \circ_{\mathrm{op}} f = (h \circ_{\mathrm{op}} g) \circ_{\mathrm{op}} f$$

and

$$g \circ_{\mathrm{op}} 1_A = 1_A \circ g = g \qquad 1_A \circ_{\mathrm{op}} f = f \circ 1_A = f$$

Definition 1.5. Let $\mathcal{C}$ be a category and $f : A \to B$ a morphism in $\mathcal{C}$. We call f an **isomorphism**, if there exists $g : B \to A$, such that $f \circ g = 1_B$ and $g \circ f = 1_A$.

We call $f : A \to B$ a **split monomorphism** if there exists $g : B \to A$, such that $g \circ f = 1_A$ and similarly f is a **split epimorphism** if there exists $g : B \to A$, such that $f \circ g = 1_B$.

Remark. Note that if f is an isomorphism, then the inverse morphism g is unique and that f is an isomorphism if and only if it is both a split epimorphism and a split monomorphism.

Definition 1.6. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A **(covariant) functor** $F : \mathcal{C} \to \mathcal{D}$ consists of

1. an assignment $\mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ taking $A \mapsto F(A)$
2. for each $A, B \in \mathrm{ob}(\mathcal{C})$ an assignment $\mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$ which takes $f \mapsto F(f)$

such that it preserves composition $F(g \circ f) = F(g) \circ F(f)$ and the identity $F(1_A) = 1_{F(A)}$.
We define a **contravariant functor** $F : \mathcal{C} \to \mathcal{D}$ to be just a covariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$.

Definition 1.7. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. We call F **faithful**, if it is injective on hom-sets. More precisely for all $A, B \in \mathrm{ob}(\mathcal{C})$ and each pair of morphisms $A \underset{g}{\overset{f}{\rightrightarrows}} B$ if $F(f) = F(g)$ then $f = g$.
We say that F is **full** if it is surjective on hom-sets. So for all $A, B \in \mathrm{ob}(C)$ and $g : F(A) \to F(B)$ there exists $f : A \to B$ such that $F(f) = g$.

Remark. Note that the definition doesn't say anything about the map on objects being injective or surjective, nor does it say that if the images of any two morphisms are equal then they are the same, since the functor may map the morphisms between two different objects to the same morphism in the target category.

1.1 Natural transformations

Definition 1.8. Let $F, G : \mathcal{C} \to \mathcal{D}$ be a pair of functors. A **natural transformation** $\eta : F \to G$ is a collection of morphisms in $\mathcal{D}$ indexed by the objects of $\mathcal{C}$, $\eta = \{\eta_A : F(A) \to G(A) \mid A \in \mathrm{ob}(\mathcal{C})\}$, such that for all objects $A, B \in \mathcal{C}$ and morphisms $f : A \to B$ the following square diagram commutes

$$\begin{tikzcd}
F(A) \arrow[r, "F(f)"] \arrow[d, "\eta_A"'] & F(B) \arrow[d, "\eta_B"'] \\
G(A) \arrow[r, "G(f)"] & G(B)
\end{tikzcd}$$

For each $A \in \mathcal{C}$ we call η_A **the component of** η **at** A. A natural transformation η is a **natural isomorphism** if for all $A \in \mathrm{ob}(\mathcal{C})$ the morphism η_A is an isomorphism.

Definition 1.9. Given two natural transformations $\alpha : F \to G$ and $\beta : G \to H$ as such

$$\begin{tikzcd}
\mathcal{C} \arrow[r, bend left=60, "F"] \arrow[r, "G"] \arrow[r, bend right=60, "H"'] & \mathcal{D}
\end{tikzcd}$$

$\Downarrow \alpha$, $\Downarrow \beta$

We define their **vertical composite** $\beta \circ \alpha$ componentwise by taking $\forall A \in \mathrm{ob}(\mathcal{C}) : (\beta \circ \alpha)_A = \beta_A \circ \alpha_A$.

Proposition 1.1. *The horizontal composite $\beta \circ \alpha : F \to H$ is a natural transformation.*

Důkaz. We need to show that the square diagram

$$\begin{tikzcd}
F(A) \arrow[r, "(\beta\circ\alpha)_A"] \arrow[d, "F(f)"] & H(A) \arrow[d, "H(f)"] \\
F(B) \arrow[r, "(\beta\circ\alpha)_B"] & H(B)
\end{tikzcd}$$

commutes. From the definition of $(\beta \circ \alpha)$ we have

$$\begin{tikzcd}
F(A) \arrow[r, "\alpha_A"] \arrow[d, "F(f)"] & G(A) \arrow[r, "\beta_A"] \arrow[d, "G(f)"] & H(A) \arrow[d, "H(f)"] \\
F(B) \arrow[r, "\alpha_B"] & G(B) \arrow[r, "\beta_B"] & H(B)
\end{tikzcd}$$

and from the definition of natural transformation it follows that the left and right squares commute, so the whole diagram commutes as well. $\square$

Definition 1.10. Let $1_F : F \to F$ be defined componentwise as $(1_F)_A = 1_{F(A)}$, then 1_F is a natural transformation which we will call the identity.

Definition 1.11. Given any two categories $\mathcal{C}, \mathcal{D}$ we define a **functor category** $[\mathcal{C}, \mathcal{D}]$ whose objects are functors from $\mathcal{C}$ to $\mathcal{D}$ and whose morphisms are natural transformations. We define the composition in $[\mathcal{C}, \mathcal{D}]$ as horizontal composition and we let the identity morphisms be the identity natural transformations.

Proposition 1.2. *$[\mathcal{C}, \mathcal{D}]$ is a category.*

Proof. Choose $A \in \mathrm{ob}(\mathcal{C})$, then

$$(\gamma \circ (\beta \circ \alpha))_A = \gamma_A \circ (\beta \circ \alpha)_A = \gamma_A \circ (\beta_A \circ \alpha_A) = (\gamma_A \circ \beta_A) \circ \alpha_A = (\gamma \circ \beta)_A \circ \alpha_A = ((\gamma \circ \beta) \circ \alpha)_A$$

$$(\alpha \circ 1_F)_A = \alpha \circ 1_{F(A)} = \alpha_A \qquad (1_F \circ \beta)_A = 1_{F(A)} \circ \beta_A = \beta_A$$

$\square$

Definition 1.12. Given any four functors and two natural transformations like so

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F_1"] \arrow[r, bend right, "G_1"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \arrow[r, bend left, "F_2"] \arrow[r, bend right, "G_2"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{E} \end{tikzcd}$$

we define **the horizontal composite** $\beta * \alpha : F_2 \circ F_1 \to G_2 \circ G_1$ by

$$\forall A \in \mathcal{C} : (\beta * \alpha)_A = \beta_{G_1(A)} \circ F_2(\alpha_A)$$

Remark. Notice that $\beta_{G_1(A)} \circ F_2(\alpha_A) = G_2(\alpha_A) \circ \beta_{F_1(A)}$ by the naturality of β at $F_1(A)$ and $G_1(A)$.

Theorem 1.1 (Middle-four interchange law)**.** *Consider the following diagram*

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F_1"] \arrow[r, bend right, "H_1"'] \arrow[r, "G_1"] \arrow[r, phantom, shift left=3, "\Downarrow\alpha_1"] \arrow[r, phantom, shift right=3, "\Downarrow\alpha_2"] & \mathcal{D} \arrow[r, bend left, "F_1"] \arrow[r, bend right, "H_2"'] \arrow[r, "G_2"] \arrow[r, phantom, shift left=3, "\Downarrow\beta_1"] \arrow[r, phantom, shift right=3, "\Downarrow\beta_2"] & \mathcal{E} \end{tikzcd}$$

then $(\beta_2 \circ \beta_1) * (\alpha_2 \circ \alpha_1) = (\beta_2 * \alpha_2) \circ (\beta_1 * \alpha_1)$.

Definition 1.13. Given a diagram like so

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \arrow[r, "H"] & \mathcal{E} \end{tikzcd}$$

we define the **whiskering of** H **and** α as a natural transformation $H\alpha : HF \to HG$ where

$$\forall A \in \mathrm{ob}(\mathcal{C}) : (H\alpha)_A = H(\alpha_A)$$

Definition 1.14. Given a diagram

$$\begin{tikzcd} \mathcal{C} \arrow[r, "F"] & \mathcal{D} \arrow[r, bend left, "G"] \arrow[r, bend right, "H"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{E} \end{tikzcd}$$

we define the **whiskering of** β **and** F as a natural transformation $\beta F : GF \to HF$ where

$$\forall A \in \mathrm{ob}(\mathcal{C}) : (\beta F)_A = \beta_{F(A)}$$

Remark.

$$H\alpha = 1_H * \alpha \qquad \beta F = \beta * 1_F$$

Definition 1.15. Given two functors $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ then the **adjunction** $F \dashv G$ is given by a pair of natural transformations $\eta : 1_{\mathcal{C}} \to GF$ called the **unit** and $\epsilon : FG \to 1_{\mathcal{D}}$ called the **counit** such that the following diagrams commute (in their respective functor categories)

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, "1_F"'] & FGF \arrow[d, "\epsilon F"] \\ & F \end{tikzcd} \qquad \begin{tikzcd} G \arrow[r, "\eta G"] \arrow[dr, "1_F"'] & GFG \arrow[d, "G\epsilon"] \\ & G \end{tikzcd}$$

Definition 1.16. Let $F : \mathcal{C} \to \mathcal{C}$ be an endofunctor, $\eta : 1_{\mathcal{C}} \to F$ and $\mu : F^2 \to F$ a pair of natural transformations. We then call the triple (F, μ, η) a **monad**, if the following two diagrams commute

$$\begin{tikzcd} F \arrow[r, "F\eta"] \arrow[dr, equal] & F^2 \arrow[d, "\mu"] & F \arrow[l, "\eta F"'] \arrow[dl, equal] \\ & F & \end{tikzcd} \qquad \begin{tikzcd} F^3 \arrow[r, "F\mu"] \arrow[d, "\mu F"'] & F^2 \arrow[d, "\mu"] \\ F^2 \arrow[r, "\mu"'] & F \end{tikzcd}$$

Definition 1.17. We call the 6-tuple $(\mathcal{C}, \otimes, I, \alpha, \rho, \lambda)$ a **monoidal category**, where $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, $I \in \mathcal{C}$ and

$$X \otimes (Y \otimes Z) \xrightarrow[\alpha_{X,Y,Z}]{\simeq} (X \otimes Y) \otimes Z$$

is a natural isomorphism natural in X, Y and Z, called the **associator** and

$$I \otimes X \xrightarrow[\lambda_X]{\simeq} X \qquad X \otimes I \xrightarrow[\rho_X]{\simeq} X$$

are a pair of natural isomorphisms, λ is called the **left unitor** and ρ is called the **right unitor** such that the following diagrams commute

$$\begin{tikzcd} & (X \otimes Y) \otimes (Z \otimes W) \arrow[dr, "\alpha_{X\otimes Y,Z,W}"] & \\ X \otimes (Y \otimes (Z \otimes W)) \arrow[ur, "\alpha_{X,Y,Z\otimes W}"] \arrow[d, "1_X \otimes \alpha_{Y,Z,W}"'] & & ((X \otimes Y) \otimes Z) \otimes W \\ X \otimes ((Y \otimes Z) \otimes W) \arrow[rr, "\alpha_{X,Y\otimes Z,W}"] & & (X \otimes (Y \otimes Z)) \otimes W \arrow[u, "\alpha_{X,Y,Z} \otimes 1_W"'] \end{tikzcd}$$

$$\begin{tikzcd} X \otimes (I \otimes Y) \arrow[rr, "\alpha_{X,I,Y}"] \arrow[dr, "1_X \otimes \lambda_Y"'] & & (X \otimes I) \otimes Y \arrow[dl, "\rho_X \otimes 1_Y"] \\ & X \otimes Y & \end{tikzcd}$$

and lastly $\lambda_I = \rho_I$.

Proposition 1.3. *Functor $F : \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if and only if F is full, faithful and essentialy surjective on objects.*

Proof. 1. F equivalence $\implies$ F faithful

Choose $f, g : A \to B$ and suppose that $F(f) = F(g)$, there exists a functor $G : \mathcal{D} \to \mathcal{C}$ and a natural isomorphism $\eta : G \circ F \to 1_{\mathcal{C}}$.

Consider the following diagram

$$\begin{tikzcd} A \arrow[drr, "f"] & & \\ G(F(A)) \arrow[u, "\eta_A"] \arrow[r, "G(F(f))=G(F(g))"'] \arrow[d, "\eta_A"'] & G(F(B)) \arrow[r, "\eta_B"] & B \\ A \arrow[urr, "g"'] & & \end{tikzcd}$$

both the upper and lower squares commute by naturality, hence $f \circ \eta_A = g \circ \eta_A \implies f = g$ since η_A is an isomorphism.

2. F equivalence $\implies$ F full

   Let $g : F(A) \to F(B)$ and define $f = \eta_B \circ G(g) \circ \eta_A^{-1}$ then $f : A \to B$. Since G is an equivalence of categories it suffices to show that $GF(f) = G(g)$ by 1. From naturality we have that

$$\begin{tikzcd} GF(A) \arrow[r, "GF(f)"] \arrow[d, "\eta_A"] & GF(B) \arrow[d, "\eta_B"] \\ A \arrow[r, "f"] & B \end{tikzcd}$$

   commutes and therefore

$$GF(f) = \eta_B^{-1} \circ f \circ \eta_A = G(g) \implies F(f) = g$$

3. F equivalence $\implies$ F essentialy surjective on objects

   Choose an object $B \in \mathrm{ob}(\mathcal{D})$, then we have $\mu_B : F(G(B)) \to B$ an isomorphism.

4. F fully faithful and essentialy surjective on objects $\implies$ F is an equivalence

   For $B \in \mathrm{ob}(\mathcal{D})$ choose $A \in \mathrm{ob}(\mathcal{C})$ such that $F(A) \cong B$ fix any such isomorphism, call it μ_B and let $G(B) = A$. Given any morphism $g : B \to C$ in $\mathcal{D}$ we have

$$\begin{tikzcd} FG(B) \arrow[r, shift left, "\mu_B"] & B \arrow[l, shift left, "\mu_B^{-1}"] \arrow[r, "g"] & C \arrow[r, shift left, "\mu_C^{-1}"] & FG(C) \arrow[l, shift left, "\mu_C"] \end{tikzcd}$$

   by full faithfulness of F there exists a unique morphism $f : G(B) \to G(C)$ such that $F(f) = \mu_C^{-1} \circ g \circ \mu_B$. We let $G(g) = f$. We will now prove, that $F(G(g_2) \circ G(g_1)) = FG(g_2 \circ g_1)$. Then it will follow by faithfulness, that G respects composition.

$$\begin{tikzcd} B \arrow[r, "g_1"] \arrow[d, shift left, "\mu_B^{-1}"] & C \arrow[r, "g_2"] \arrow[d, shift left, "\mu_C^{-1}"] & D \arrow[d, shift left, "\mu_D^{-1}"] \\ FG(B) \arrow[u, shift left, "\mu_B"] & FG(C) \arrow[u, shift left, "\mu_C"] & FG(D) \arrow[u, shift left, "\mu_D"] \end{tikzcd}$$

   We have $F(G(g_2)G(g_1)) = FG(g_2)FG(g_1) = \mu_3 g_2 \mu_2^{-1} \mu_2 g_1 \mu_1 = \mu_3 g_2 g_1 \mu_1 = FG(g_2 \circ g_1)$ and $FG(1_B) = 1_{FG(B)} = F(1_{G(B)})$. Now we have to check, that $FGB \cong B$ naturally in B. This follows easily, we define a natural transformation $\mu : FG \to 1_{\mathcal{D}}$ whose components are those that we have picked and the naturality square

$$\begin{tikzcd} FGB \arrow[r, "FGg"] \arrow[d, "\mu_B"] & FGC \arrow[d, "\mu_C"] \\ B \arrow[r, "g"'] & C \end{tikzcd}$$

   commutes by the definition of $G(g)$ and μ is a natural isomorphism. Choose $A \in \mathrm{ob}(\mathcal{C})$, there exists $A' \in \mathrm{ob}(\mathcal{C})$ such that

$$\begin{tikzcd} F(A) \arrow[r, shift left, "\mu_{F(A)}"] & F(A') \arrow[l, shift left, "\mu_{F(A)}^{-1}"] \end{tikzcd}$$

   thus $GF(A) = A'$. From full faithfulness of F, there exists an isomorphism $\eta_A : A \to A'$ such that $F(\eta_A) = \mu_{F(A)}$. Define $\eta : 1_{\mathcal{C}} \to GF$, whose components are η_A. Consider

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\eta_A"] & B \arrow[d, "\eta_B"] \\ GFA \arrow[r, "GF(f)"] & GFB \end{tikzcd}$$

where $GF(f)$ is the unique morphism such that $FGF(f) = \mu_{F(B)}^{-1} \circ F(f) \circ \mu_{F(A)}$. Therefore the F image of this diagram commutes and by faithfulness of F the square above commutes as well, which completes the proof.

□

1.2 Limits and colimits

Definition 1.18. Let $\mathcal{J}$ be a small category. A **diagram** D **of shape** $\mathcal{J}$ is any functor $D : \mathcal{J} \to \mathcal{C}$. Define $\Delta : \mathcal{C} \to [\mathcal{J}, \mathcal{C}]$ which sends A to the constant functor ΔA that assigns all $j \in \mathrm{ob}(\mathcal{J})$ to A and every morphism in $\mathcal{J}$ to the identity morphism 1_A. Given $A \in \mathrm{ob}(\mathcal{C})$ a **cone with vertex** A is a pair (A, λ), where $\lambda : \Delta A \to D$ is a natural transformation. A **universal cone** is a cone (A, λ) so that for any other cone (B, μ) there exists a unique morphism $f : B \to A$ such that $\forall j \in \mathcal{J} : \lambda_j \circ f = \mu_j$.

Lemma 1.1. *Universal cone of a diagram D is unique up to a unique isomorphism, if it exists.*

Proof. □

Definition 1.19. The **limit of a diagram** D is its universal cone, if it exists.

Lemma 1.2 (Pullback lemma)**.** *Consider this commutative diagram*

$$\begin{tikzcd}
\bullet \arrow[r, "f_1"'] \arrow[d, "f_2"] & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
\bullet \arrow[r, "p_2"'] & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

assuming that the right square is a pullback, then the left square is a pullback iff the outer rectangle is a pullback.

Proof. Assume that the outer rectangle is a pullback and consider any cone (U, s_1, s_2) like so

$$\begin{tikzcd}
U \arrow[rr, bend left, "s_1"] \arrow[dr, bend right, "s_2"'] & & & \\
& \bullet \arrow[r, "f_1"'] \arrow[d, "f_2"] & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "p_2"'] & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

since the outer rectangle is a pullback there exists a unique morphism m that makes the triangles in the following diagram commute

$$\begin{tikzcd}
U \arrow[dr, "m"] \arrow[rr, bend left, "f_3 \circ s_1"] \arrow[dr, bend right, "s_2"'] & & \\
& \bullet \arrow[r, "f_3 \circ f_1"'] \arrow[d, "f_2"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "q_2 \circ p_2"'] & \bullet
\end{tikzcd}$$

because $(U, f_3 \circ s_1, s_2)$ is a cone above the outer rectangle. Now we have to prove that $f_1 \circ m = s_1$

$$\begin{tikzcd}
U \arrow[dr, "m"] \arrow[rr, bend left, "s_1"] \arrow[dr, bend right, "s_2"'] & & & \\
& \bullet \arrow[r, "f_1"'] \arrow[d, "f_2"] & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "p_2"'] & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}
\qquad
\begin{tikzcd}
U \arrow[dr, "s_1"] \arrow[dr, bend left, "f_1 \circ m"] \arrow[rr, bend left, "f_3 \circ s_1"] \arrow[dr, bend right, "p_1 \circ f_1 \circ m"'] & & \\
& \bullet \arrow[r, "f_3"] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

By a diagram chase we verify, that the outer square on the right side really commutes. From the universal property of pullbacks we get that $f_1 \circ m = s_1$ as required. Now to show that m is unique consider the commutative diagram

$$\begin{tikzcd}
U \arrow[dr, "m'"] \arrow[ddr, bend right, "s_2"'] \arrow[drr, bend left, "s_1 \circ f_3"] & & \\
& \bullet \arrow[r, "f_3 \circ f_1"'] \arrow[d, "f_2"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "q_2 \circ p_2"'] & \bullet
\end{tikzcd}$$

it follows by the universal property that $m = m'$, since the outer rectangle is a pullback. Now assume that the left square is a pullback and consider any cone (U, s_1, s_2)

$$\begin{tikzcd}
U \arrow[ddr, bend right, "s_2"'] \arrow[drrr, bend left, "s_1"] & & & \\
& \bullet \arrow[r, "f_1"'] \arrow[d, "f_2"] & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "p_2"'] & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

Since $(U, s_1, p_2 \circ s_2)$ is a cone and the right square is a pullback, there exists unique n such that $f_3 \circ n = s_1$ and $p_1 \circ n = p_2 \circ s_2$, but then (U, s_2, n) is a cone on the left pullback square, so there exists a unique m such that $f_1 \circ m = n$ and $f_2 \circ m = s_2$ like so

$$\begin{tikzcd}
U \arrow[dr, "m"] \arrow[drr, bend left, "n"] \arrow[ddr, bend right, "s_2"'] \arrow[drrr, bend left, "s_1"] & & & \\
& \bullet \arrow[r, "f_1"'] \arrow[d, "f_2"] & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "p_2"'] & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

simple diagram chase shows that m completes the cone (U, s_1, s_2). It remains to show that m is unique. Let m' be any other completion of the cone (U, s_1, s_2)

$$\begin{tikzcd}
U \arrow[dr, "m'"] \arrow[drr, bend left, "n"] \arrow[ddr, bend right, "s_2"'] \arrow[drrr, bend left, "s_1"] & & & \\
& \bullet \arrow[r, "f_1"'] \arrow[d, "f_2"] & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& \bullet \arrow[r, "p_2"'] & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

if we show $f_1 \circ m' = n$ then by the universal property we would have that $m = m'$. By a diagram chase we see that $(U, s_1, p_2 \circ s_2)$ is a cone completed by n and $f_1 \circ m'$ of the right pullback square

$$\begin{tikzcd}
U \arrow[drr, bend left, "n"] \arrow[drr, bend right, "f_1 \circ m'"'] \arrow[ddr, bend right, "p_2 \circ s_2"'] \arrow[drrr, bend left, "s_1"] & & & \\
& & \bullet \arrow[r, "f_3"'] \arrow[d, "p_1"] & \bullet \arrow[d, "q_1"] \\
& & \bullet \arrow[r, "q_2"'] & \bullet
\end{tikzcd}$$

therefore from universality $f_1 \circ m' = n$ and again by universality of the left pullback $m = m'$. $\square$

Definition 1.20. Let $\mathcal{C}$ be a category and choose $X \in \mathrm{ob}(\mathcal{C})$. We define the **slice category $\mathcal{C}$ over** X denoted as $\mathcal{C}/X$, whose objects are pairs (A, f) where $A \in \mathrm{ob}(\mathcal{C})$, $f : A \to X$ and whose morphisms $g : (A, f_1) \to (B, f_2)$ are just morphisms $g : A \to B$ in $\mathcal{C}$ such that the following diagram commutes

$$\begin{tikzcd} A \arrow[rr, "g"] \arrow[dr, "f_1"'] & & B \arrow[dl, "f_2"] \\ & X & \end{tikzcd}$$

Notice that composition of two morphisms $\mathcal{C}/X$ yields again a morphisms in $\mathcal{C}/X$ since when the two triangles in the left diagram below commute then the entire square commutes as well and that by the very definition of identity morphism the triangle on the right side below commutes

$$\begin{tikzcd} & B \arrow[dr, "g_2"] \arrow[dd, "f_2"] & \\ A \arrow[ur, "g_1"] \arrow[dr, "f_1"'] & & C \arrow[dl, "f_3"] \\ & X & \end{tikzcd} \qquad \begin{tikzcd} A \arrow[rr, "1_A"] \arrow[dr, "f_1"'] & & A \arrow[dl, "f_1"] \\ & X & \end{tikzcd}$$

Definition 1.21. Given two objects A, B their coproduct is an object U with two morphisms $\iota_A : A \to U$ and $\iota_B : B \to U$, such that for any other triple $(U', f : A \to U', g : A \to U')$ there exists a unique morphism $f + g : U \to U'$, such that the following diagram commutes

$$\begin{tikzcd} & U' & \\ A \arrow[ur, "f"] \arrow[r, "\iota_A"] & U \arrow[u, dashed, "f+g"] & B \arrow[ul, "g"'] \arrow[l, "\iota_B"'] \end{tikzcd}$$

Proposition 1.4. *Coproduct U is unique up to an isomorphism.*

1.3 Extensivity

A category is extensive if it has all finite coproducts and for all $X_1, X_2 \in \mathcal{C}$ the functor

$$F_+ : \mathcal{C}/X_1 \times \mathcal{C}/X_2 \to \mathcal{C}/(X_1 + X_2)$$

is an equivalence of categories. We define F_+ as follows

$$F_+((A_1, f_1 : A_1 \to X_1), (A_2, f_2 : A_2 \to X_2)) = (A_1 + A_2, (\iota_{X_1} \circ f_1) + (\iota_{X_2} \circ f_2))$$

where $(\iota_{X_1} \circ f_1) + (\iota_{X_2} \circ f_2)$ is the unique map (existence and uniqueness follows from universal property of $A_1 + A_2$) such that the following diagram commutes

$$\begin{tikzcd} A_1 \arrow[r, "\iota_{A_1}"] \arrow[d, "f_1"'] & A_1 + A_2 \arrow[d, dashed, "\exists!"] & A_2 \arrow[l, "\iota_{A_2}"'] \arrow[d, "f_2"] \\ X_1 \arrow[r, "\iota_{X_1}"'] & X_1 + X_2 & X_2 \arrow[l, "\iota_{X_2}"] \end{tikzcd}$$

Given a morphism in $\mathcal{C}/X_1 \times \mathcal{C}/X_2$

$$(h_1, h_2) : ((A_1, f_1 : A_1 \to X_1), (A_2, f_2 : A_2 \to X_2)) \to ((B_1, g_1 : B_1 \to X_1), (B_2, g_2 : B_2 \to X_2))$$

so $h_1 : A_1 \to B_1 \wedge h_2 : A_2 \to B_2$ such that the following two triangles commute

$$\begin{tikzcd} A_1 \arrow[r, "h_1"] \arrow[dr, "f_1"'] & B_1 \arrow[d, "g_1"] \\ & X_1 \end{tikzcd} \qquad \begin{tikzcd} A_2 \arrow[r, "h_2"] \arrow[dr, "f_2"'] & B_2 \arrow[d, "g_2"] \\ & C_2 \end{tikzcd}$$

we define $F_+(h_1, h_2)$ as the unique morphism $(\iota_{B_1} \circ h_1) + (\iota_{B_2} \circ h_2)$ which make the following diagram commute

$$\begin{tikzcd}
A_1 \arrow[r, "\iota_{A_1}"] \arrow[d, "h_1"] & A_1 + A_2 \arrow[d, dashed, "\exists!"] & A_2 \arrow[l, "\iota_{A_2}"'] \arrow[d, "h_2"] \\
B_1 \arrow[r, "\iota_{B_1}"] & B_1 + B_2 & B_2 \arrow[l, "\iota_{B_2}"']
\end{tikzcd}$$

Now to show that it is a morphism in the slice category, we have to verify that the following triangle commutes

$$\begin{tikzcd}
A_1 + A_2 \arrow[r, "(\iota_{B_1} \circ h_1) + (\iota_{B_2} \circ h_2)"] \arrow[dr, "(\iota_{X_1} \circ f_1) + (\iota_{X_2} \circ f_2)"'] & B_1 + B_2 \arrow[d, "(\iota_{X_1} \circ g_1) + (\iota_{X_2} \circ g_1)"] \\
& X_1 + X_2
\end{tikzcd}$$

Lemma 1.3. *Morphism from coproduct is uniquely determined by composition with its insertions. Consider the following diagram*

$$\begin{tikzcd}
X_1 \arrow[r, "\iota_{X_1}"] \arrow[dr, "g \circ \iota_{X_1} = f \circ \iota_{X_1}"'] & X_1 + X_2 \arrow[d, shift right, "g"'] \arrow[d, shift left, "f"] & X_2 \arrow[l, "\iota_{X_2}"'] \arrow[dl, "f \circ \iota_{X_2} = g \circ \iota_{X_2}"] \\
& X &
\end{tikzcd}$$

then $f = g$.

Proof. Follows directly from the universal property of coproducts that there exists a unique arrow $\kappa : X_1 \times X_2 \to X$ that makes the triangles commute, but both f and g make the triangles commute, so $f = \kappa = g$. □

Consider the diagram

$$\begin{tikzcd}
A_1 \arrow[r, "\iota_{A_1}"] \arrow[d, "f_1"] \arrow[dd, bend right=60, "h_1"'] & A_1 + A_2 \arrow[d] \arrow[dd, bend left=60] & A_2 \arrow[l, "\iota_{A_2}"'] \arrow[d, "f_2"'] \arrow[dd, bend right=60, "h_2"] \\
B_1 \arrow[r, "\iota_{B_1}"] \arrow[d, "g_1"] & B_1 + B_2 \arrow[d] & B_2 \arrow[l, "\iota_{B_2}"'] \arrow[d, "g_2"'] \\
X_1 \arrow[r, "\iota_{X_1}"] & X_1 + X_2 & X_2 \arrow[l, "\iota_{X_2}"']
\end{tikzcd}$$

To prove that the inner triangle (whose unmarked arrows are as above) commutes we chase around the diagram starting from the insertions by the lemma above. We know by assumption that the leftmost and rightmost triangles commute and that all the four black squares commute by definition of the unmarked arrows and the same for the two red squares.

Now we ought to show that F_+ really is a functor. We have

$$F_+(1_{A_1}, 1_{A_2}) = (\iota_{A_1} \circ 1_{A_1}) + (\iota_{A_2} \circ 1_{A_2}) = \iota_{A_1} + \iota_{A_2} = 1_{A_1 + A_2} = 1_{F_+((A_1, f_1), (A_2, f_2))}$$

the penultimate equality again follows by the universal property of $A_1 + A_2$.

Given two morphisms $(h_1 : A_1 \to B_1, h_2 : A_2 \to B_2), (k_1 : B_1 \to C_1, k_2 : B_2 \to C_2)$ we have

$$F_+((k_1, k_2) \circ (h_1, h_2)) = F_+((k_1 \circ h_1, k_2 \circ h_2)) = (\iota_{C_1} \circ k_1 \circ h_1) + (\iota_{C_2} \circ k_2 \circ h_2)$$

Notice that to prove this equals the desired

$$((\iota_{C_1} \circ k_1) + (\iota_{C_2} \circ k_2)) \circ ((\iota_{B_1} \circ h_1) + (\iota_{B_2} \circ h_2))$$

amounts to showing that the unmarked inner triangle in the following diagram commutes provided that the four square diagrams commute (the square once again commute because of the universality of coproduct) and that the two big rectangles commute

$$\begin{tikzcd}
A_1 \arrow[r, "\iota_{A_1}"] \arrow[d, "h_1"'] & A_1 + A_2 \arrow[d] \arrow[dd, bend left] & A_2 \arrow[l, "\iota_{A_2}"'] \arrow[d, "k_1"] \\
B_1 \arrow[r, "\iota_{B_1}"] \arrow[d, "h_2"'] & B_1 + B_2 \arrow[d] & B_2 \arrow[l, "\iota_{B_2}"'] \arrow[d, "k_2"] \\
C_1 \arrow[r, "\iota_{C_1}"] & C_1 + C_2 & C_2 \arrow[l, "\iota_{C_2}"']
\end{tikzcd}$$

we again make use of the lemma and chase around the diagram until we get the equality. It follows, that $F_+((k_1, k_2) \circ (h_1, h_2)) = F_+((k_1, k_2)) \circ F_+(h_1, h_2)$.

Proposition 1.5. *A category with finite coproducts is extensive if and only if it has pullbacks along all insertions of binary coproducts and given that the following diagram commutes*

$$\begin{tikzcd}
A_1 \arrow[r, "\omega_1"] \arrow[d, "f_1"] & A \arrow[d, "f"] & A_2 \arrow[l, "\omega_2"'] \arrow[d, "f_2"] \\
X_1 \arrow[r, "\iota_{X_1}"] & X_1 + X_2 & X_2 \arrow[l, "\iota_{X_2}"']
\end{tikzcd}$$

then A_1 and A_2 are a pair of pullbacks if and only if A is a coproduct of A_1 and A_2 in the category of $\mathcal{C}/(X_1 + X_2)$,ω_1, ω_2 are insertions and $f = f_1 + f_2$.

Proof. □

2 Monoids and acts

Definition 2.1. A 3-tuple $\mathcal{S} = (S, \star, e)$ where S is a set and $\star : S \times S \to S$, $e : 1 \to S$ are maps is called a **monoid**, if the following two conditions hold

1. $\forall x, y, z \in S : (x \star y) \star z = x \star (y \star z)$
2. $\forall x \in S : x \star e = x = e \star x$

Example 2.1. *Let X be a nonempty set. Denote by $T(X)$ the set of all maps $X \to X$.*
Then $\mathcal{T}(X) = (T(X), \circ, id_X)$, where $\circ$ is a composition of maps and id_X the identity map, is a monoid. We will call $\mathcal{T}(X)$ the transformation monoid of X.

Let $\mathcal{S} = (S, \star, e)$ be a monoid, we define the opposite monoid $\mathcal{S}^{op} = (S, \star_{op}, e)$ where

$$\forall x, y \in S : x \star_{op} y = y \star x$$

Definition 2.2. Let $\mathcal{S}$ be a monoid and X a nonempty set. A tuple (X, μ), where $\mu : S \times X \to X$ is called a **left $\mathcal{S}$-act** if the map μ satisfies the following

1. $\forall r, s \in S \, \forall x \in X : \mu(rs, x) = \mu(r, \mu(s, x))$
2. $\forall x \in S : \mu(e, x) = x$

we will denote a left $\mathcal{S}$-act as ${}_{\mathcal{S}}X$. Similarly we define a right $\mathcal{S}$-act and use the notation $X_{\mathcal{S}}$.

Remark. When we use the word $\mathcal{S}$-act we will mean the left $\mathcal{S}$-act, unless stated otherwise.

Definition 2.3. Let X be a set and $\mathcal{S}$ a monoid. We define the **free $\mathcal{S}$-act on X** to be the tuple

$$(S \times X, \cdot : S \times (S \times X) \to S \times X)$$

where $s_1 \cdot (s, x) = (s_1 s, x)$.

It is an act, since $e \cdot (s, x) = (es, x) = (s, x)$ and $s_1 s_2 \cdot (s, x) = ((s_1 s_2)s, x) = (s_1(s_2 s), x) = s_1 \cdot (s_2 s, x) = s_1 \cdot (s_2 \cdot (s, x))$.

Definition 2.4. Given ${}_{\mathcal{S}}X$ and ${}_{\mathcal{S}}Y$ we call $f : X \to Y$ an **act homomorphism**, if

$$\forall s \in S \, \forall x \in X : f(s \cdot x) = s \cdot f(x)$$

Proposition 2.1. *Acts with their homomorphisms form a category, which we will denote $\mathcal{S}$-**Act**.*

Construction 2.1 (Powerset Act Monad)**.** *Consider an act ${}_{\mathcal{S}}A$, then we have an endofunctor $\mathcal{P}$ in $\mathcal{S}$-**Act**.*

$${}_{\mathcal{S}}A \mapsto_{\mathcal{S}} P(A)$$

given

3 Universal Algebra

Definition 3.1. We call a tuple $(\Omega, \tau : \Omega \to \mathbb{N}_0)$, where Ω is a set of symbols and τ is a so called **arity function** a **type**. We will often times identify the type with its underlying set Ω, when the arity function is implied.

Definition 3.2. We call the tuple $(X, \{F_\omega\}_{\omega \in \Omega})$, where X is a set and $F_\omega : X^{\tau(\omega)} \to X$ are maps indexed by Ω, an **algebra of type** Ω.

Definition 3.3. Let $(X, \{F_\omega\}_{\omega \in \Omega})$ and $(Y, \{G_\omega\}_{\omega \in \Omega})$ be algebras of type Ω, then $\phi : X \to Y$ is a **Ω-homomorphism** if

$$\forall \omega \in \Omega \, \forall \{x_i\}_{i=1}^{\tau(\omega)} \subseteq X : \phi(F_\omega(x_1, \ldots, x_{\tau(\omega)})) = G_\omega(\phi(x_1), \ldots, \phi(x_{\tau(\omega)}))$$

Proposition 3.1. *The identity $1_X : X \to X$ is an Ω-homomorphism and the composition of two Ω-homomorphism yields a Ω-homomorphism.*

Proposition 3.2. *Ω-algebras with their homomorphisms form a category, which we will call* $\mathbf{Alg}_\Omega$.

Definition 3.4. Let $(X, \{F_\omega\}_{\omega\in\Omega})$ be an Ω-algebra and ρ an equivalence relation on X. Then we call ρ a **congruence on** X, if or each $\omega \in \Omega$ we have

$$\forall\{x_i\}_{i=1}^{\tau(\omega)}\,\forall\{y_i\}_{i=1}^{\tau(\omega)} : x_1 \sim y_1 \wedge \cdots \wedge x_{\tau(\omega)} \sim y_{\tau(\omega)} \implies F_\omega(x_1, \ldots, x_{\tau(\omega)}) \sim F_\omega(y_1, \ldots, y_{\tau(\omega)})$$

Proposition 3.3. *Given an Ω-homomorphism $\phi : X \to Y$ we have that the relation*

$$\ker\phi = \{(x, y) \in X^2 : \phi(x) = \phi(y)\}$$

is a congruence relation on X.

Definition 3.5. Given $\sim$ a congruence on Ω-algebra $(X, \{F_\omega\}_{\omega\in\Omega})$ then we define the **quotient Ω-algebra of X by** $\sim$ as

$$(X/\sim, \{H_\omega\}_{\omega\in\Omega})$$

where $X/\sim$ is the set of equivalence classes of X by $\sim$ and H_ω is defined as

$$H_\omega([x_1]_\sim, \ldots, [x_{\tau(\omega)}]_\sim) = [F_\omega(x_1, \ldots, x_{\tau(\omega)})]_\sim$$

Remark. Well-definedness of the maps follows from the defining property of congruences.

Definition 3.6. We define the **quotient map** $\pi_\sim : X \to X/\sim$ as $x \mapsto [x]_\sim$.

Remark. It is straightforward to check that the quotient map is an Ω-homomorphism.

Proposition 3.4. *Given $f : X \to Y$ and $\sim \leq \ker\phi$ a congruence on X, there exists a unique morphism $f^\sim$, such that the following triangle commutes*

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "\pi_\sim"'] & & Y \\ & X/\sim \arrow[ur, "f^\sim"'] & \end{tikzcd}$$

Proof. Let $f^\sim([x]_\sim) = f(x)$. The uniqueness is trivial. □

Given any $Y \in \mathbf{Alg}_\Omega$ we have a endofunctor in the slice category $F : \mathbf{Alg}_\Omega/Y \to \mathbf{Alg}_\Omega/Y$

$$(X, \phi : X \to Y) \mapsto (X/\ker(\phi), \phi^{\ker(\phi)})$$

$$f : (X_1, \phi_1) \to (X_2, \phi_2) \mapsto F(f) : F(X_1) \to F(X_2)$$

where

$$Ff : [x]_{\ker(\phi_1)} \mapsto [f(x)]_{\ker(\phi_2)}$$

In fact it may be a monad

Definition 3.7. Let $(X_i, \{F^i_\omega\}_{\omega\in\Omega}), i \in I$ be a family of Ω-algebras indexed by I. Then their product is an Ω-algebra given by

$$\left(\prod_{i\in I} X_i, \{F_\omega\}_{\omega\in\Omega}\right)$$

where for $i \in I$ and $f_1, \ldots, f_{\tau(\omega)} \in \prod_{i\in I} X_i$ we have

$$F_\omega(f_1, \ldots, f_{\tau(\omega)})(i) = F^i_\omega(f_1(i), \ldots, f_{\tau(\omega)}(i))$$

For $j \in I$ we define the j-th projection $p_j : \prod_{i\in I} X_i \to X_j$ as

$$p_j(f) = f(j)$$

Remark. The projections are homomorphisms, since

$$p_j(F_\omega(f_1, \ldots, f_{\tau(\omega)})) = F_\omega(f_1, \ldots, f_{\tau(\omega)})(j) = F^j_\omega(f_1(j), \ldots, f_{\tau(\omega)}(j)) = F^j_\omega(p_j(f_1), \ldots, p_j(f_{\tau(\omega)}))$$

Proposition 3.5. *The category* $\mathbf{Alg}_\Omega$ *admits products.*

Proof. Given any family $X_i, i \in I$ assume there is an object Q with morphisms $q_i : Q \to X_i$ for each $i \in I$. □